

DANIEL SIMBOLON

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EXECUTIVE SUMMARY

I am an Electrical Engineer and a graduate of Del Institute of Technology with experience in electrical systems, instrumentation, and automation. I am accustomed to working in teams and leading on-site activities, including transmitter installation, PLC upgrades, and CCTV system installation. I have also served as a Site Manager on several projects, responsible for team coordination, technical supervision, and safety compliance. With strong skills in electronics, IoT, microcontrollers, PCB design, and prototyping, I am adaptable, solution-oriented, and committed to delivering valuable results.

EDUCATION

Del Institute of Technology – Sumatera Utara

Sep 2020 – Aug 2024

Bachelor degree (S-1) in Electrical Engineering, 3.28/4.00

Thesis : Design and Development of an Electric Coffee Bean Roasting Machine.

Relevant Courses : Control Systems, Embedded Systems, Internet of Things 2020– 2024.

EXPERIENCE

PT THOPAS ARTHA NAULI

Jakarta Selatan, DKI Jakarta

Electrical Engineer

Jan 2025 - Present

Principal Responsibilities:

- Installation, wiring, and commissioning of instrumentation systems.
- PLC configuration, upgrade, and troubleshooting for industrial automation systems.
- CCTV system installation and integration, including power supply, networking, and functional testing.
- On-site coordination and technical supervision as Site Manager in compliance with safety and technical standards.
- Technical support to customers through site visits, technical presentations, specification reviews, and solution recommendations.

PT ATMI MIKAEL FORTUNA

Bekasi, Jawa Barat

Spesial Purpose Machine

Aug 2023- Jun 2024

Principal Responsibilities:

- Create 3D CAD designs for efficient and compliant coffee machines.
- Design and install electrical systems ensuring safety standards.
- Select quality materials and components for durability and performance.
- Perform QC testing to ensure optimal machine performance.

PT ATMI MIKAEL FORTUNA

Bekasi, Jawa Barat

Research and Development

Jun 2023 – Aug 2023

Principal Responsibilities:

- Select a battery type considering capacity, voltage, and power needs.
- Design series-parallel circuits to meet device power requirements.
- Integrate the battery pack with a BMS for monitoring and protection.
- Add a fastcharging module for efficient and safe charging ports.
- Design a protective minimalist battery cover.
- Test charging, discharging, and ports for safety and performance.

PROJECT

Electrical & PLC Engineer – I/O Upgrade & Panel Modification Project

Sep 2025 - Oct 2025

Principal Responsibilities:

- Supervised and executed PLC I/O module upgrades based on system requirements and site conditions.
- Performed replacement and rearrangement of PLC I/O and control panel components to improve system reliability.
- Verified compliance between engineering drawings/wiring diagrams and actual field implementation.
- Updated and redrew control panel layout (as-built drawings) to reflect the final configuration.
- Coordinated wiring, labeling, and panel workmanship in accordance with industry standards.

Site Manager Electrical & Instrumentation Installation Project

Jun 2025 – Aug 2025

Principal Responsibilities:

- Supervised CCTV (Low Voltage System) installation to ensure technical compliance and proper security coverage.
- Managed site work permits and enforced HSE compliance throughout project execution.
- Provided technical coordination and on-site involvement in CCTV installation, including installation verification and functional testing.
- Supervised installation of Rosemount magnetic flow meters (Instrumentation) in accordance with vendor installation guidelines.
- Coordinated mechanical and electrical works, including flange alignment, power supply, grounding, and signal cable routing.

Design and Development of an Electric Coffee Bean Roasting Machine.

Aug 2023 - Jul 2024

Principal Responsibilities:

- Identify and determine the appropriate materials and technical components, such as drum roasting materials, heating elements, and blowers, to support the optimal coffee roasting process.
- Design and install an integrated electrical system, including heating element connections, motor drives, and temperature sensors, to ensure safe and efficient machine performance.
- Calculate and set the optimal drum drive motor speed to ensure even heat distribution during the coffee roasting process.

Battery Pack System Design and Development.

Jun 2023 - Aug 2023

Principal Responsibilities:

- Designing and Making Battery Pack with Fast Charging Output System
- Integrating Battery Pack with Battery Management System (BMS) Module

Solar-Powered smart irrigation and fertilization with loRa remote monitoring

Mar 2023 – Jun 2023

Principal Responsibilities:

- Design and Implement a Routine Irrigation and Fertilization System
- Installing Integrated Irrigation System Wiring with Solar Panels
- Conducting Functional Testing and Collecting Irrigation Data for 6 Months

PCB Design for Audio Amplifier

Sep 2023 - Oct 2023

Principal Responsibilities:

- Create a circuit design that integrates components such as transistors, op-amps, or amplifier ICs to boost audio signals from input sources.
- Create an optimal PCB layout to support the performance of the audio amplifier circuit, including the arrangement of signal paths, ground, and power supplies.
- Test the amplifier's function to ensure proper signal enhancement, good sound quality, and compatibility with source and output audio devices.

Indonesian Robot Contes

Apr 2022 – Jul 2022

Principal Responsibilities:

- Designing and Building a Material Handling Robot with a Claw System Driven by Remote Control
- Developing an Arduino Program to Implement Robot Control Functions, Including Claw Movement and Navigation
- Selecting and Integrating Optimal Electronic Components, Such as DC Motors, Servos, and Remote Modules, to Improve Robot Performance
- Conducting Testing and Validating Remote Control Functions to Ensure Responsiveness and Accuracy of Robot Movement

Smart Fire Alarm System with BD139 Transistor Technology

Januari 2022

Principal Responsibilities:

- Conducting electronic circuit design to detect potential fire using BD139 transistor as the main component.
- Creating a simple PCB layout that is efficient and suitable to support the integration of a fire alarm system.
- Testing the function of ensuring the fire alarm circuit in various scenarios to clarify and the system's responsibility for fire conditions.

Innovative Arduino-Controlled Automatic Hand Washing System

October 2021

Principal Responsibilities:

- Designed a hand detection system using an HC-SR04 sensor and Arduino to control air and soap flow based on hand distance.
- Developed Arduino code to automate solenoid valves and pumps, ensuring efficient and responsive system performance.

ORGANIZATIONAL EXPERIENCE

- **Himpunan Mahasiswa Elektro, Del Institute of Technology** *Member* *Oct 2020 – Jul 2022*
- **Del Robotic Club, Del Institute of Technology** *Head of Facilities and Infrastructure Division* *Feb 2021 – Jun 2022*
- **Himpunan Mahasiswa Elektro, Del Institute of Technology** *Head of Community Service Department* *Jul 2022 – Jul 2023*
- **Del Robotic Club, Del Institute of Technology** *Chairman* *Jun 2022 – Aug 2023*
- **Ikatan Alumni Elektro, Del Institute of Technology** *Chairman of the 2020 Electrical Engineering alumni* *Oct 2024 – Present*

TECHNICAL SKILLS

- **Language skills** : Bahasa Indonesia (Native) | English (Intermediate)
- **Skills**
 - Microsoft Word : Provide a report with charts, symbols, tables, and pictures.
 - Microsoft Excel : Provide a report with charts, symbols, tables, pictures, and data calculations.
 - Power Point : Provide and present a report with video, pictures, charts, and an amazing slide show.
- **Soft Skills** : Leadership, Public Speaking, Time Management, Hardworking, Discipline, Creativity, Analytical Thinking, Problem Solving, Teamwork, Event Management, Adaptability, Presentations
- **Programming Language** : Python, C, C++, HTML, Ladder Diagram
- **Tools** : Arduino IDE, FluidSIM, Visual Studio Code, Eagle, Solidworks, Proteus, Multisim, Ultimaker Cura, Ms. Office, Notepad++, Eagle CX-One, CX-Programmer.

PUBLICATIONS

- **Solar-Powered smart irrigation and fertilization with loRa remote monitoring** *August 2023*
Certificate: <https://www.idss.iocspublisher.org/index.php/jidss/article/view/145/95>
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CERTIFICATE

- **PLC Basic Certification Training from Omron** *August 2024*
Certificate: <https://bit.ly/3DJQ4aS>
- **Internship Participant** *August 2023*
Certificate: <https://bit.ly/3XT0ziR>
- **Participant of the Indonesian Robot Contest 2022 (KRI 2022) Merdeka Campus with a focus on the KRTMI Robot Contest (Indonesian Thematic Robot Contest) Indonesia** *June 2022*
Certificate: <https://bit.ly/3D1CFzW>
- **Finalists of E-NNOVATE (International Innovation Show) organized by the Association of Young Indonesian Scientists** *April 2022*
Certificate: <https://bit.ly/427NrZC>
- **Finalists of the 2022 USM International Virtual Fair organized by the Indonesian Young Scientists Association** *April 2021*
Certificate: <https://bit.ly/3XRNJ4I>